

## Mamba2: A Linear Attention Layer

The quadratic  $\mathcal{O}(\text{seqlen}^2)$  scaling of the standard Transformers attention mechanism has increasingly become a bottleneck as context lengths have ballooned with the advent of reasoning models and rising prevalence of modalities such as video and audio.

Mamba2 [1] is an alternative information propagation algorithm belonging to the steadily growing class of *linear*  $\mathcal{O}(\text{seqlen})$  attention mechanisms. Its GPU-aware design enables hardware utilization comparable to quadratic attention during training, and reduces the decoding time and cache-space from  $\mathcal{O}(\text{seqlen})$  to  $\mathcal{O}(1)$ . [GG: MODEL REFS!](#) [GG: Arch summary table](#)

Fully leveraging this improved scaling requires efficient training of Mamba2-based models on long-sequence documents, which poses engineering challenges due to the linear memory growth with context length. Context-parallelism (CP), in which sequences are sharded along the sequence dimension across GPUs, is a natural and scalable approach to long-sequence training. This poster describes context-parallel implementation for Mamba2. [GG: Should probably mention ring attn somewhere](#)



Figure 1: Pink-necked green pigeon [2].

## Mamba2 Architecture (Simplified)

Mamba2 relies on two central mechanisms for propagating information:

1. Short 1D causal convolutions [GG: github link](#)
2. A gated recursion relation

Letting  $x_{sd} \in \mathbb{R}^{\text{seqlen} \times \text{d\_model}}$  be an input tensor, the Mamba2 outputs are of the schematic form

$$z_{sd} \sim \text{gated\_recursion}(\text{causal\_conv1d}(x_{sd}))$$

Both components, described in some more detail below, require adaptation in the context-parallel implementation. For brevity, we suppress batch and head dimensions throughout.

### Causal Convolutions

The causal convolution is a depthwise, 1D convolution along the sequence dimension with a short filter, typically of width  $K = 4$  [3]:

$$z_{sd} = \sum_{k=0}^K W_{dk} x_{(s-k)d} \cdot$$

### Gated Recursion Relations

The central elements in the Mamba2 recursion relation are of the form:

$$z_{sd} = e^{-A_s} z_{(s-1)d} + \Delta_s x_{sd}$$

where the data-dependent  $A_s, \Delta_s \geq 0$  control the deletion and addition of information to the state  $z_{sd}$ . While the complete tensor  $z_{sd}$  can be constructed in  $\mathcal{O}(\text{seqlen})$  by solving the recursion in the naive manner, such an approach is suboptimal in practice as it cannot leverage GPU tensor cores. For this reason, the recursion is solved using a chunked, matmul-based strategy which has inferior theoretical, big- $\mathcal{O}$  scaling, but superior in-practice wall times [1].

[GG: figs](#)

## Context Parallel Convolutions

[GG: fig](#)

Context-parallel causal convolutions can be implemented with minimal changes [GG: edit language](#). For `seqlen` tokens split into chunks of length  $C = \text{seqlen} / \text{cp\_degree}$ , a naive intra-chunk convolution whose outputs are nearly all correct: only the first  $K - 1 \ll C$  tokens require fixing.

An efficient algorithm is a follows:

1. CP rank `r` asynchronously passes its final  $K - 1$  tokens (`x_send = x[-K+1:]`) to rank `r+1`
2. Concurrently, each rank runs the convolution on its locally-available tokens, producing `z = causal_conv1d(x)`
3. After the  $K - 1$  passed tokens are received into the buffer `x_rcv`, they are concatenated with the first  $K - 1$  local tokens and the leading  $K - 1$  tokens of `z` are overwritten with their their resulting convolution: `z[:K-1] = causal_conv1d(cat(x_rcv, x[:K-1]))`.

## Context Parallel Gated Recursion

[GG: fig](#)

We have implemented several context-parallel version of the Mamba2 gated recursion relation.

### Simple State Passing

One implementation leverages the fact that computing the outputs  $z_{sd}$  for  $s \geq s'$  only requires knowing the inputs  $x_{sd}$  at these token positions and the final state at this boundary, i.e.  $z_{sd}$  at  $s = s' - 1$ . The algorithm:

1. CP rank  $r = 0$  solves the recursion relation using the locally available inputs and passes its final state `z[-1]` to rank  $r = 1$ .
2. CP rank  $r = 1$  solves its recursion relation with the non-trivial initial state received from rank  $r = 0$  and passes its final state to  $r = 2$
3. Continue this pattern for all ranks.

The strict causal dependencies of the Mamba2 algorithm results in idles GPUs in this strategy, with the last CP rank experiencing  $\mathcal{O}(\text{seqlen})$  exposed communication time. If there are no synchronization points (e.g. as induced by tensor-parallelism or FSDP), it is possible to amortize these costs across successive layers via pipelining [GG: fig](#), but such sync points are common in practice.

### Compute-Then-Correct

Alternatively, a strategy similar to the CP convolution algorithm is also possible, in which we compute incorrect outputs with locally available tensors, and then correct the results via communication. GPUs never idle with this strategy.

In order to describe this strategy, we trade the global sequence index  $s$  for the pair of indices  $r, c$  with  $r \in \{0, \dots, \text{cp\_degree} - 1\}$  indexing the rank and  $c \in \{0, \dots, \text{seqlen} / \text{cp\_degree} - 1\}$  indexing the chunked sequence position. A valid CP implementation is as follows:

1. Every rank computes  $\Sigma_r = \sum_c A_{rc}$ : the sum of local gate values
2. Rank  $r$  asynchronously sends  $\Sigma_r$  to CP ranks  $r' > r$
3. Every rank solves the recursion relation with its local inputs  $x_{rcd}$  and trivial initial state, producing (incorrect) final states  $y_{rd}^{\text{final}}$
4. Rank  $r$  sends  $y_{rd}^{\text{final}}$  to CP ranks  $r' > r$
5. Rank  $r$  computes its corrected initial state via  $x_{rd}^{\text{initial}} = \sum_{r' < r} \exp(\Sigma_{r-1} + \dots + \Sigma_{r'+1}) y_{r'd}^{\text{final}}$
6. Every rank re-solves the recursion relation with its now-corrected initial states, yielding the correct outputs  $z_{rcd}$

This strategy requires  $\mathcal{O}(\text{seqlen} / \text{cp\_degree})$  additional compute and has  $\mathcal{O}(\text{cp\_degree})$  exposed communication.



Biological Information

|            |                |
|------------|----------------|
| Domain     | Eukaryota      |
| Kingdom    | Animalia       |
| Phylum     | Chordata       |
| Class      | Aves           |
| Clade      | Columbimorphae |
| Order      | Columbiformes  |
| Family     | Columbidae     |
| Type genus | Columba        |

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#pop.update-poster-layout(...)
#pop.update-theme()
```

Peace of Posters Documentation

You can find more information on the documentation site under [jonaspleyer.github.io/peace-of-posters/](https://jonaspleyer.github.io/peace-of-posters/).

Figure 2: The poster from the thumbnail can be viewed at the documentation website as well.

General Relativity

Einstein’s brilliant theory of general relativity starts with the field equations [4].

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$$

However, they have nothing to do with doves.

Peace be with you

Figure 3: ‘Doves [...] are used in many settings as symbols of peace, freedom or love. Doves appear in the symbolism of Judaism, Christianity, Islam and paganism, and of both military and pacifist groups.’ [5].

Etymology

Pigeon is a French word that derives from the Latin pīpiō, for a ‘peeping’ chick, while dove is an ultimately Germanic word, possibly referring to the bird’s diving flight. The English dialectal word culver appears to derive from Latin columba [6]. A group of doves is called a “dule”, taken from the French word deuil (‘mourning’) [7].

References

[1]

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[4]

A. Einstein, “Die Grundlage der allgemeinen Relativitätstheorie [AdP 49, 769 (1916)],” *Annalen der Physik*, vol. 517, no. S1, pp. 517–571, 1916, doi: 10.1002/andp.2005517s151.

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